

Aymen Sadraoui, Ph.D.

Data Scientist & Data Engineer

Machine Learning, Clinical AI & Big Data Systems

Paris, France

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[Portfolio](#)

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Professional Summary

I am a **Data Scientist** and **Machine Learning Ph.D.** graduate specialized in building robust **predictive models** and scalable **data engineering pipelines** for high-dimensional, multimodal datasets. I bring extensive hands-on expertise in **deep learning**, **representation learning**, and **few-shot learning**, maintaining a rigorous evaluation-first methodology focused on **model calibration**, **out-of-distribution generalization**, and reproducibility. Proven track record of translating complex mathematical research into first-author publications at top-tier venues (**CVPR**, **ISBI**, **ICIP**) and engineering production-grade **ETL pipelines** for clinical AI systems. I am seeking roles where I can architect end-to-end **data pipelines** and deploy scalable, impact-driven **AI/ML solutions**.

Professional Experience

Data Scientist & Machine Learning Engineer

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Feb 2026 – Present

Paris, France

- Architected and deployed end-to-end **automated data processing and ingestion pipelines** integrating directly with hospital PACS (DICOM) and cloud platforms for multi-centric clinical studies.
- Developed deep learning-based algorithms for **automated image quality control (QC)**, detecting motion artifacts, slice mismatches, and data corruption prior to clinical review.
- Engineered computer vision algorithms for **automated diagnosis, segmentation, and classification** of radiology scans (CT, X-ray, MRI) across orthopedics, oncology, and rheumatology modalities.
- Deployed scalable, containerized microservices (**Docker**, **FastAPI**, **PyTorch**) into production workflows, powering AI-driven pre-annotations and interactive 3D visualizations for clinical operators.
- Implemented robust **MLOps frameworks** for model calibration, stress testing, and data drift detection to ensure high out-of-distribution generalization across multi-site hospital data.
- Partnered with cross-functional teams of radiologists, software engineers, and clinical managers to translate clinical research requirements into production-ready software features.

Ph.D. Researcher in AI & Computational Pathology

CentraleSupélec (OPIS / CVN Lab) & Kremlin-Bicêtre Hospital

Jan 2022 – Jan 2026

Gif-sur-Yvette, France

- Thesis:** *AI-Driven Methods in Histopathology for the Diagnosis and Prognosis of Hepatocellular Carcinoma.*
- Clinical-Technical Interface:** Served as the primary bridge between ML researchers and senior pathologists; translated qualitative histopathological criteria into quantifiable ML objectives and validated automated features against clinical consensus.
- Advanced Deep Learning & Optimization:** Designed and unrolled proximal optimization algorithms (Unrolled Projected Gradient, UNEM) into neural networks for robust stain separation, color normalization, and transductive few-shot slide classification.
- Terabyte-Scale Data Engineering:** Built scalable ETL preprocessing pipelines (PyTorch, OpenCV, OpenSlide, Ray) to ingest, tile, and curate over 4+ TB of gigapixel whole slide images, handling high-throughput data processing and multi-source artifact removal.
- Multi-Modal Predictive Modeling:** Developed deep learning models integrating histopathologic architecture quantification with clinical/prognostic clinical metadata to predict early liver cancer recurrence.
- MLOps & Trustworthy AI:** Created rigorous benchmarking frameworks focused on out-of-distribution generalization, model calibration, uncertainty estimation, and interpretability for high-stakes medical deployment.

Clinical AI & ML Engineering Intern

CentraleSupélec & SUP'COM

Feb 2021 – Sep 2021

France & Tunisia

- Implemented deep neural network pipelines for digital liver histopathology slides and representation learning for hepatocellular carcinoma classification.
- Developed end-to-end data ingestion and image-processing pipelines for real-world environmental data analysis.

Technical Skills

Languages & Core Tech: Python, SQL, Bash/Linux, Git, CI/CD

Data Engineering & Infrastructure: Docker, Kubernetes, Optuna, ETL/ELT Pipelines, Data Modeling

Machine Learning & Deep Learning: PyTorch, Scikit-learn, OpenCV, OpenSlide, Multimodal Modeling, Representation Learning, Few-Shot Learning, Convex Optimization, Computer Vision, NLP

MLOps & Model Deployment: Model Calibration, Robustness & Stress Testing, Out-of-Distribution Generalization, MLflow, FastAPI, Docker, Experiment Tracking, Data Drift Detection

Languages: English (Fluent), French (Fluent), Arabic (Native)

Achievements

- **1st Place Winner:** Lung Transplant Rejection Challenge – Organized by Hôpital Foch & Health Data Hub (2024)
- **National Rank 63 / 1,773 (Top 3%):** National Engineering Entrance Exam, Tunisia (2018)

Education

Ph.D. in Machine Learning & Artificial Intelligence <i>CentraleSupélec, Université Paris-Saclay</i>	Jan 2022 – Jan 2026 <i>Gif-sur-Yvette, France</i>
Engineering Degree in Applied Mathematics & Computer Science <i>École Polytechnique de Tunisie</i>	2018 – 2021 <i>La Marsa, Tunisia</i>
Preparatory Classes (Mathematics & Physics) <i>Preparatory Institute for Engineering Studies</i>	2016 – 2018 <i>Bizerte, Tunisia</i>

Publications

- **Sadraoui, A., et al.** (2026). A Proximal Approach for Stain Separation and Normalization of Whole-Slide Histopathological Images. *IEEE 23rd International Symposium on Biomedical Imaging (ISBI)*.
- **Sadraoui, A., et al.** (2025). UNEM: UNrolled Generalized EM for Transductive Few-Shot Learning. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- **Sadraoui, A., et al.** (2024). Deep Learning Classification and Quantification of Pejorative and Nonpejorative Architectures in Resected Hepatocellular Carcinoma from Digital Histopathologic Images. *The American Journal of Pathology*
- **Sadraoui, A., et al.** (2024). Unrolled Projected Gradient Algorithm for Stain Separation in Digital Histopathological Images. *IEEE International Conference on Image Processing (ICIP)*
- **Sadraoui, A., et al.** (2024). A Transductive Few-Shot Learning Approach for Classification of Digital Histopathological Slides from Liver Cancer. *IEEE International Symposium on Biomedical Imaging (ISBI)*